

# Keystroke Dynamics Against Academic Dishonesty in the Age of LLMs

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## Abstract

*The transition to online examinations and assignments raises significant concerns about academic integrity. Traditional plagiarism detection systems often struggle to identify instances of intelligent cheating, particularly when students utilize advanced generative AI tools to craft their responses. This study proposes a keystroke dynamics-based method to differentiate between bona fide and assisted writing within academic contexts. To facilitate this, a dataset was developed to capture the keystroke patterns of individuals engaged in writing tasks, both with and without the assistance of generative AI. The detector, trained using a modified TypeNet architecture, achieved accuracies ranging from 74.98% to 85.72% in condition-specific scenarios and from 52.24% to 80.54% in condition-agnostic scenarios. The findings highlight significant differences in keystroke dynamics between genuine and assisted writing. The outcomes of this study enhance our understanding of how users interact with generative AI and have implications for improving the reliability of digital educational platforms.*

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## 1. Introduction

With the rapid advancements in generative AI technology and increased reliance on virtual educational environments, cheating in online examinations has significantly escalated. Many of us can now confidently identify AI-generated plagiarism, yet it remains a challenging task, especially in large class settings. Testing agencies have implemented several online proctoring measures to uphold integrity during examinations. These include prohibiting tab changes, disabling the copy-paste function, and preventing the minimization of the exam window. Such restrictions aim to mimic the controlled environment of in-person exams and ensure a level playing field. However, these measures fall short of addressing all forms of academic dishonesty. Traditional plagiarism detection tools like *Turnitin* and *Urkund* work by comparing submissions against existing sources. While these tools effectively detect

direct plagiarism, they often fail to recognize more sophisticated cheating methods, such as extensive paraphrasing or external assistance via generative AI.

Figure 1 illustrates a framework diagram for existing systems that process a primary input, typically a query or suspicious document ( $Q_d$ ), and an optional reference collection, such as the web ( $D$ ). These systems output identified suspicious fragments and their sources of plagiarism ( $S_q$ ) when available. However, they are ineffective against more sophisticated forms of cheating, including text manipulation, extensive paraphrasing, and assistance from another individual [1]. The emergence of advanced large language models like ChatGPT and Gemini has further complicated the landscape. These AI systems can generate high-quality, human-like essays within seconds, posing new challenges for current plagiarism detection mechanisms. The capability of these tools to produce tailored, undetectable responses is particularly problematic for traditional plagiarism detection software, which relies on direct text comparison.

Numerous studies have explored plagiarism within academic circles, outlining various types of plagiarism and potential detection methods [1–4]. Linguistic analyses have indicated that identifying specific contexts is crucial for effective plagiarism detection [5, 6]. For instance, in introductory computing courses, cheating involving ChatGPT can be detected by comparing student code submissions with those generated by ChatGPT, achieving an accuracy of 90% or better. Notable differences often include conciseness, reduced variable duplication, simpler control flow, and a preference for ternary operators in the code produced by ChatGPT [5].

The ability to distinguish between AI-generated and human-generated text has garnered significant interest from a wide range of researchers since the introduction of ChatGPT [7–13]. While traditional natural language processing techniques concentrated on the content of responses, recent studies show that behavioral signals, from the text’s production (such as typing behavior) to its consumption (including reading or gaze behavior), offer valuable insights about the text [14, 15]. Typing behavior captured by keystroke dynamics provides extensive information about text production processes. This method has become a well-established, non-intrusive behavioral modality for user authentication, which we are adapting to detect plagiarism [16–21].

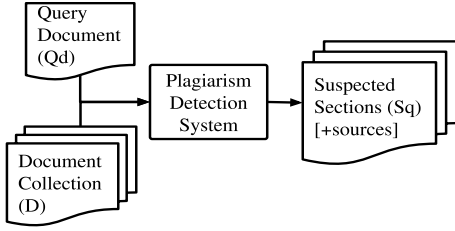


Figure 1. The framework of existing plagiarism detectors such as Turnitin and Urkund, which use content-based word-for-word similarities [1, 22].

The Educational Testing Service (ETS) has effectively utilized keystroke log features to predict essay scores. Models specific to tasks, genres, and contexts, trained on features such as time spent on tasks, burst lengths in words and pauses within words, have demonstrated good generalizability across various prompts [23].

Another prevalent form of academic dishonesty in online or distance learning examinations is impersonation, often referred to as contract cheating [24]. Keystroke analysis can address this issue by comparing the keystroke dynamics of the registered examinee with those submitted during the examination[25,26].

Previous research in writing cognition, including Hayes and Flower’s model [27], identified key sub-processes in writing: idea generation, translation of ideas into text, typing, and revising. Figure 2 depicts a simplified version of Hayes’ cognitive writing model [28], which includes four sub-processes: *Proposer*, *Translator*, *Transcriber*, and *Evaluator*. These models suggest that authentic writing tasks require more planning and revising than assisted writing, highlighting the importance of these processes in effective writing. Inspired by previous studies [28–31], we hypothesize and investigate whether keystroke dynamics and linguistic analyses can capture the distinctive cognitive processes behind bona fide and assisted writing.

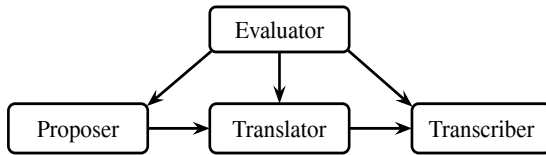


Figure 2. The cognitive model of the writing process includes four stages: *Proposer*, where ideas are generated and tasks prepared, marked by initial pauses and pauses at sentence boundaries; *Translator*, involving the fluent conversion of ideas into language, measured by uninterrupted text production sequences; *Transcriber*, focusing on orthographic proficiency and motor skills, evident in pauses within words and immediate spelling corrections; and *Evaluator*, which entails editing and reviewing, identifiable by jumps to different text parts for extensive edits [30].

In other words, the cognitive stages of the writing process are closely linked to distinctive keystroke dynamics, providing a window into the writer’s cognitive states. For example, the *Proposer* stage could be characterized by initial pauses and pauses at sentence boundaries, reflecting the cognitive load of idea

generation. In the *Translator* stage, fluent typing with minimal interruptions indicates efficient translation of thoughts to text. The *Transcriber* stage is characterized by pauses within words and immediate corrections, highlighting the focus on spelling and grammatical accuracy. Finally, the *Evaluator* stage involves extensive cursor movements and edits across the text, indicative of thorough reviewing and editing processes. We can infer whether a text was produced by following a natural writing process or plagiarized by analyzing keystroke patterns—like pause lengths, typing rhythm, and editing behaviors.

In essence, our goal is to train and evaluate a suite of classifiers—denoted as  $f_K$  for keystroke dynamics  $X_K$  to detect academic dishonesty by classifying text and corresponding keystrokes submitted by students as bona fide ( $y = 0$ ) or AI-assisted ( $y = 1$ ), across various academic tasks and contexts. The evaluation is conducted under diverse training and testing setups to explore the impact of task complexity and contextual variations on classifier performance. In summary, our main contributions are as follows:

- The study introduces a keystroke dynamics-based method for detecting academic dishonesty, employing the TypeNet model to differentiate between bona fide and AI-assisted writing.
- It proposes a unique dataset capturing keystroke patterns during various writing tasks for training and evaluating the plagiarism detection models.
- Conducts a comprehensive evaluation that reveals the method’s effectiveness under different scenarios, including variations across keyboards, users, and contexts, while also discussing the inherent challenges and limitations in typing behavior analysis.
- The paper highlights the potential of keystroke dynamics as a supplementary tool to traditional plagiarism detection methods and proposes future research directions to enhance algorithm accuracy and ensure fairness in academic assessments.

The remainder of the paper is organized as follows: Section 2 discusses related work; Section 3 outlines the experimental design; Section 4 presents the results and discussion; Section 5 enumerates the limitations; and Section 6 concludes the study, offering possible future directions.

## 2. Related work

The persistent challenge of academic dishonesty, specifically plagiarism, has spurred numerous technological advancements aimed at detection and prevention. Historically, tools like Turnitin have relied on textual analysis to detect similarities between submitted work and existing materials [32]. Despite these efforts, students have found ways to bypass such systems using sophisticated methods such as paraphrasing or employing ghostwriters. Consequently, recent studies have adopted machine learning approaches, notably autoencoders, and LSTM networks, to analyze patterns in student assessment data. Kamalov et al. [33] highlighted the efficacy of combining LSTMs with kernel density estimation for detecting discrepancies in student performance, which is crucial for identifying instances

of academic dishonesty during post-exam analyses. Additionally, surveillance technologies, including video monitoring and IP address tracking, have enhanced the integrity of examinations [34, 35].

Keystroke dynamics, which record neuro-physiological characteristics comparable to handwritten signatures, have been widely studied. In the domain of continuous user authentication, these dynamics are studied to assess latency times between keystrokes, thus enhancing password security and thwarting unauthorized access [21, 36, 37]. The applications of keystroke dynamics also extend to uncovering the anonymity of online content, automating the estimation of soft biometrics [38], detecting lies [39], identifying deceptive reviews [40], and discerning fake profiles on social media [40–42]. Additionally, keystroke analysis is crucial for authorship attribution and profiling, enabling the prediction of demographic characteristics such as age and gender based on typing patterns [43, 44]. Importantly, Goodman et al. [45] have applied stylometric features with raw keystroke data to precisely predict the authors of emails, thereby confirming the accuracy of keystroke dynamics in authorship attribution.

The inherent capabilities of keystroke dynamics in authorship attribution and user authentication indicate their potential applicability for plagiarism detection. The accurate measurement of typing rhythms and patterns may reveal the authenticity of the text production process. For instance, abrupt changes in typing speed or key pressure could signify a transition from original writing to copied text, thereby serving as indicators of plagiarism. This method could significantly aid traditional content-based plagiarism detection systems because it is agnostic to language and content, concentrating instead on the behavioral aspects exhibited during writing.

While keystroke dynamics have demonstrated efficacy in diverse fields, their application in plagiarism detection remains underexplored. This study addresses this gap by employing keystroke dynamics to distinguish between authentic and AI-assisted or plagiarized texts. Focusing on typing behavior under various cognitive loads, this research broadens the application of keystroke dynamics beyond simple authentication and authorship attribution, transforming them into a valuable tool for upholding academic integrity. The proposed models are designed to detect subtle variations in typing patterns and linguistic styles, presenting a novel method for identifying plagiarism in academic environments.

### 3. Design of experiment

The proposed framework, depicted in Figure 3, outlines the experimental setup. Detailed descriptions of each component are provided in the following sections. The dataset and implementation details are made available via Github<sup>1</sup>.

#### 3.1. Datasets

We utilize three distinct keystroke datasets, including two established collections from Stony Brook University (SBU) and SUNY Buffalo, alongside one custom dataset developed specifically for this study.

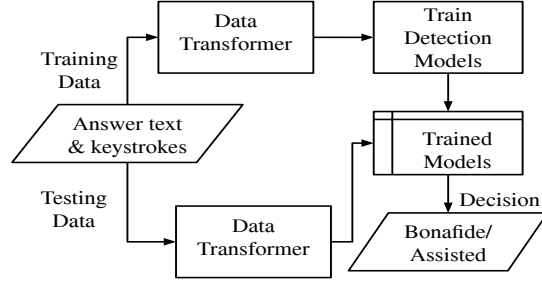


Figure 3. Illustrating the proposed detection framework that transforms training and testing data for use in training the detection models, which then contribute to deciding whether the output is bonafide or assisted.

The SBU [40] and Buffalo [46] datasets were selected as they aligned with the experimental scenarios we intend to explore. Table 1 presents a comparative analysis of their data collection processes.

**The SBU dataset** comprises both truthful and deceptive writings across two domains: restaurant business reviews and social issue essays on gun control and gay marriage. Participants were instructed to type freely and retype their entries, facilitating the analysis of natural and controlled typing behaviors, thus simulating bona fide and assisted typing conditions, respectively.

**The Buffalo dataset** contains keystroke data from 157 participants who transcribed fixed texts and responded to questions freely across three sessions using four different keyboards. This dataset is particularly valuable for examining how various keyboards affect typing behavior. This dataset’s fixed text transcription component mimics assisted typing scenarios, where participants replicate given texts, resembling situations where content is mindlessly copied.

Table 1. Details of SBU, Buffalo, and the proposed datasets.

| Characteristics       | SBU | Buffalo | Proposed |
|-----------------------|-----|---------|----------|
| Users                 | 196 | 148     | 40       |
| Free writing          | ✓   | ✓       | ✓        |
| Fixed writing         | ✓   | ✓       | ✓        |
| Keyboard labels       | ✗   | ✓       | ✗        |
| Cognitive-information | ✗   | ✗       | ✓        |
| AI-assisted writing   | ✗   | ✗       | ✓        |

**Proposed dataset** While the SBU and Buffalo datasets align well with our research goals, they do not fully encompass the complexity of keystroke dynamics in controlled academic environments, particularly in scenarios involving bona fide and AI-assisted writing. An IRB-approved data collection protocol was established to replicate these scenarios in online examination settings.

The proposed dataset comprised 40 students from various academic disciplines at a major, research-intensive university predominantly within STEM fields. This diversity ensured a comprehensive representation of typing skills and cognitive approaches. Each participant completed two separate sessions:

<sup>1</sup><https://github.com/ijcb-2024/keystroke-llm-plagiarism>